



From Data to Diagnosis: AI in Emergency Room Training

Private · Accessible · Knowledge-Grounded Medical Consultations via AI Avatar

Abhiram Varma Nandimandalam

Abhighyan Bommerla

Rishinath Reddy Enugala

Usha Goud Gourigari

Matthew Qi Lan Thompson

Instructor:

Dr. Nitika Sharma

| Advisor:

Ash Black

| In collaboration with

CoDIRA

College of Information Science · Capstone Spring 2026

01 - PROBLEM & MOTIVATION

The Rural Healthcare Crisis



60M+

Rural Americans
Underserved



21%

Lack Broadband
Access



40mi

Avg. Distance
to Hospital



High

Mental Health
Stigma Barrier



Delayed diagnoses → worse outcomes

02 - APPROACH / METHODOLOGY

RAG-Augmented Consultation Pipeline

1

User approaches kiosk → voice or text input

2

Whisper ASR transcribes speech to text

3

RAG retrieves relevant medical knowledge base docs

4

LLM generates grounded response with persona styling

5

AI avatar delivers answer via real-time video + voice

6

Quiz engine generates MCQs for self-assessment

03 - PRODUCT FEATURES

Core Capabilities



RAG Medical Chatbot

Source-cited answers from a curated medical corpus



AI Doctor Avatar

Lip-synced HeyGen video avatar



Voice Interaction

Hands-free voice consultation



Quiz Self-Assessment

Auto-generated MCQs for patient & trainee learning

04 - SYSTEM ARCHITECTURE

End-to-End Data Flow

KIOSK UI

Razor Pages
Voice Input
Avatar Display
Quiz Interface

API LAYER

ChatController
AvatarStreamCtrl
QuizController
HealthCheck API

AI SERVICES

RAGService
PersonalityLayer
WhisperASR
CharacterGateway

EXTERNALS

OpenAI GPT-4o
HeyGen Stream
Ollama (Local)
Knowledge Base



Chat + RAG



Avatar Stream



Quiz Engine



Knowledge Base



Health Checks

05 - TECHNOLOGY STACK

Tools & Frameworks

AI SERVICES

OpenAI GPT-4o · HeyGen ·
Whisper ASR · Ollama · RAG / FastAPI

FRAMEWORKS

.NET 8 · Razor Pages · REST APIs ·
Docker

LANGUAGES

C# / .NET · Python · HTML/CSS/JS

Deployment

- ▶ Containerized with Docker
- ▶ Offline-capable via Ollama local LLM
- ▶ Kiosk mode — no personal device needed
- ▶ Lightweight UI for low-resource hardware

06 - KEY RESULTS & DEMO

Performance Highlights

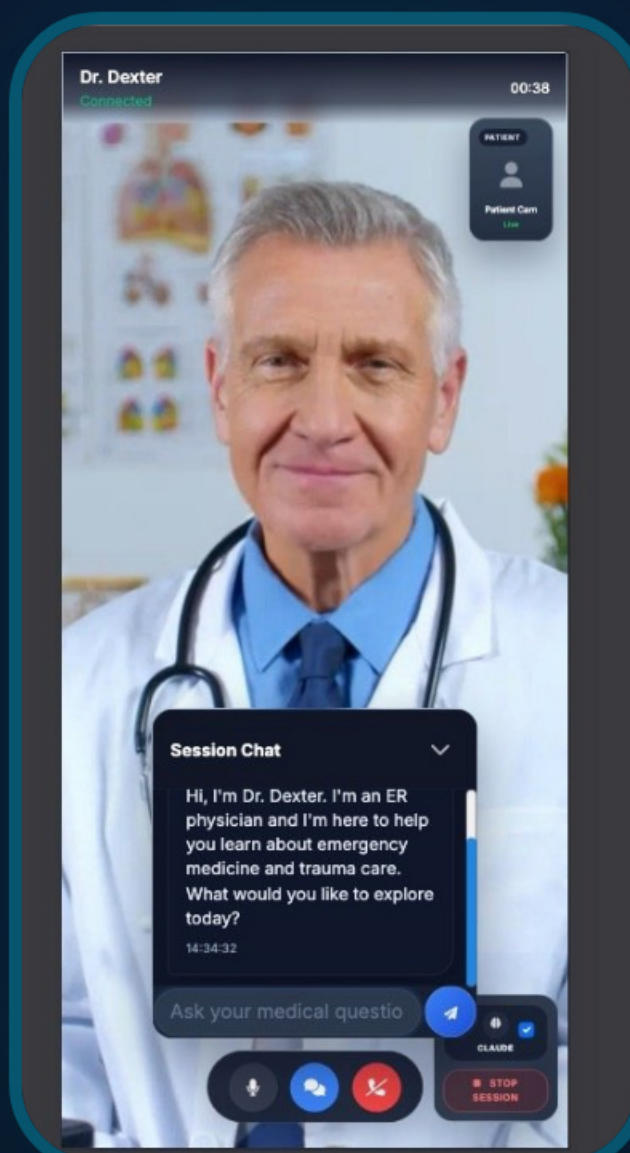
4

AI PIPELINES

<3s

AVG LATENCY

LIVE DEMO PREVIEW



Speak naturally → avatar
responds in real-time



Switch between Doctor &
Patient roles



Generate quiz on any health
topic, on demand

07 - CONCLUSION & IMPACT

Real-World Value

- ▶ Removes geographic + broadband barriers to care
- ▶ Privacy-first design reduces stigma for sensitive topics
- ▶ Multilingual, disability-first delivery — accessible to all
- ▶ Kiosk model — deploy in clinics, libraries, pharmacies



github.com/UA-AICore/Emergency-Room-Simulator



Data Privacy



Bias Mitigation



Safety Escalation



Accessibility



Transparency